# <u>FYP Proposal Document</u>

# CyberShield: Cybersecurity Awareness & Incident Reporting Portal

Final Year Project
Proposal by



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## Abstract

Phishing and social engineering attacks have increased the threat level in Pakistan as its society has grown increasingly reliant on digital platforms, particularly amongst its educational institutions and the public at large. Publics who by and large lack formal situational awareness training. To address this we propose **CyberShield**, a Multilingual Cyber Security Awareness Gateway, combining learning modules, quizzes, certificates, gamification, and hands on phishing simulations (via email, Whatsapp, and voice). Platform and role-based access for admin and user supports scalable management for institutions and universities, and offers tailored learning pathways based on performance and risk. Live reporting, AI-based phishing campaigns to make sure users have theoretical and practical understanding. Manager dashboards and analytics ensure managers can monitor As students benefit from certificates, badges and a leaderboard, CyberShield reinforces awareness and resilience to evolving cyber attacks through the fusion of education, simulation and risk analysis.

## Introduction

Cybersecurity has become one of the major concerns in the digital era. Among various hacking techniques that have been recently employed, phishing and social engineering have been recognized as the main ways through which hackers can gain illegal access to other users. Owing to the rapid digitalization in education and other sectors, the awareness and preventive measures have not kept pace with the digital exploitation in Pakistan. Consequently, students, teachers, and the civilian population are extremely exposed.

The following platform promises to close the gap by providing not only multilingual training, phishing simulations, risk-based analysis, gamified learning, role-based access for institutions, and easy incident reporting that gives power to educational sectors and the community to boost their cyber resilience but also offer these services in multiple languages, simulate phishing, analyze risks, offer gamified learning, provide role-based access for institutions, and allow quick incident reporting, thereby not only empowering the educational sectors but also the community at large to raise the level of their cyber resilience.

## Problem Statement

### Identified Problem

Pakistan has a rapidly growing cybersecurity threat. The cases of phishing and social engineering have been on the rise so that phishing attempts in 2024 have gone up by **18%** [1]. A Visa survey indicates that **91%** of users totally disregard the fraud warnings, and **52%** of them have already been deceived [2]. Poor digital awareness makes students, teachers, and the public highly vulnerable. The current platforms are enterprise-centric and lack localization, practicalization, and adaptive learning.

It is essential to have a security awareness platform that is localized and specially designed for the Pakistani education sector as well as for the general public. Such a solution would consist of multilingual training, phishing simulations, risk scoring, gamification, role-based access, and reporting in order to overcome the big difference between knowledge and actual-world readiness.

**End Customers and Targeted Audience**

**Educational Institutions (Schools, Colleges, Universities)**

- Such organizations are at the risk of data breaches as they deal with sensitive information about students and staff, and usually have a lack of cybersecurity training resources.
- Market Size: It is a large field that includes both public and private schools of all educational levels, which is one of the biggest segments due to the ongoing need for awareness among students, teachers, and school administrators.

**General Public / Individual Learners**

- Regular internet users that may be students, parents or workers not belonging to a certain organization who are in need of security awareness to keep their personal data and online identities safe.
- Market Size: This is the biggest, as the target audience covers all internet users or users of digital services, thus being the most extensive segment.

## Literature Review

Pakistan is facing a rapid spurt of cyber incidents, still the defense mechanisms and awareness about cybercrime are not only limited but also fragmented. In fact, according to the PTA and NR3C reports, Pakistan was ranked 7th globally in the list of countries with the most victims of online cybercrime in 2020, where over 100,000 cybercrime complaints were officially registered in 2022 alone [3]. The main perpetrators of the most common threats to the people, businesses, and educational institutions are financial fraud, digitally facilitated phishing scams, identity theft, and social engineering attacks.

The digital environment of Pakistan presents rapid expansion but insufficient cybersecurity measures which creates substantial dangers for its users. The fast-growing digital economy of Pakistan brought 65 million users to its internet platform in 2019 yet held the 67th position among 193 nations in the Global Cybersecurity Index revealing major security weaknesses [4]. The existing security gaps have catalyzed the rise of various cyber threats which include social media intimidation and financial deceptions and banking frauds like the 2017 ATM skimming attack against Habib Bank.

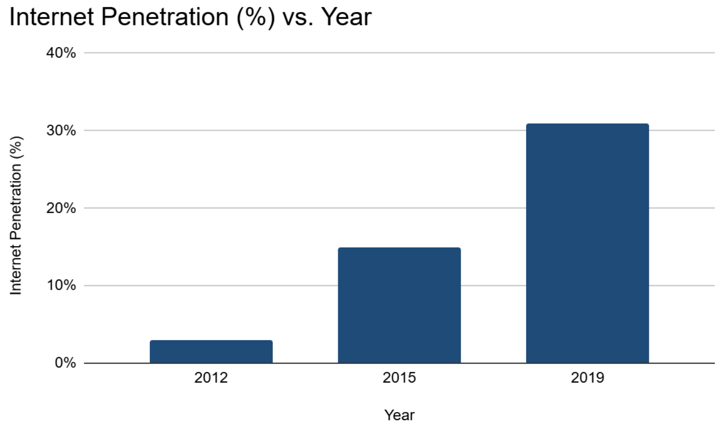

Figure 1.1: Growth of Internet Penetration in Pakistan (2012–2019)

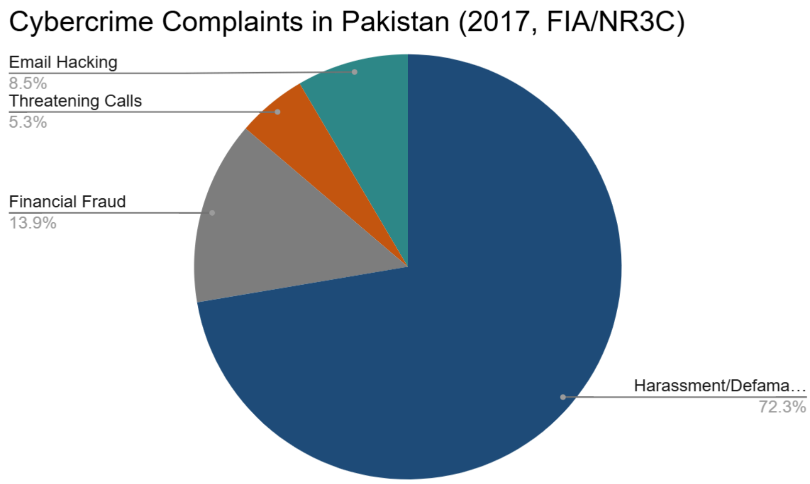


Figure 1.2: Distribution of Cybercrime Complaints Reported to FIA/NR3C in Pakistan (2017)

Pakistan saw a startling increase in cyberattacks in 2024, mostly aimed at the financial industry. Banking and financial malware attacks climbed 114% from 2023, but spyware incidents went up 63%, therefore fueling grave worries about data theft and privacy. Web-based attacks like phishing and dangerous websites struck 13.7% of Pakistan's users during the third quarter of 2024; 18.7% faced local threats distributed via CDs, USB drives, and infected installations. Industrial systems too came under intense attack; malware including trojans, backdoors, and keyloggers affected 29.51% of ICS computers in vital industries like energy and manufacturing.[5] These statistics show Pakistan's pressing need for better cybersecurity measures.

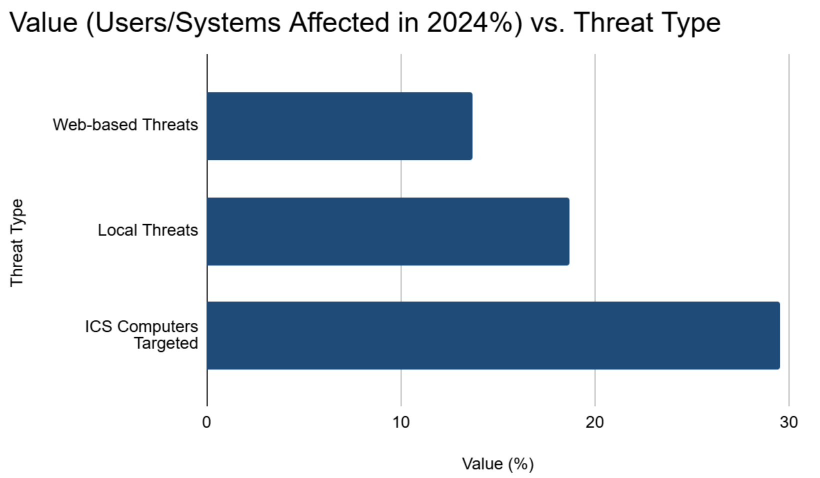


Figure 1.3: Percentage of Users/Systems Affected by Different Threat Types (2024)

Pakistan's current training and simulation platforms show two main issues: they use imported systems which do not address local requirements and their narrow coverage includes only email threats but excludes Whatsapp-smishing and voice-based phishing which are prevalent in the region. The lack of Urdu content and non-adaptive learning methods decreases their effectiveness on various audience groups including schools and universities.

A context-aware multilingual cybersecurity awareness platform needs immediate attention because of these observed gaps.

## Existing Cybersecurity Systems

- **CaniPhish:** Caniphish has a training library comprising both free and paid courses while availing awareness on phishing through configured training. Admins also run phishing campaigns using customizable email templates, phishing websites, and employee onboarding. Besides a decent mixture of learning and simulation, the course feature is majorly confined to email phishing, without adequate multilingual support and adaptation to regional contexts like Pakistani organizations or institutions, which renders it less relatable in its training. Also, training modules only allow MCQs and true/false questions, which, therefore, lacks variety and depth of assessments as they do not accommodate short or open-ended types of questions. Caniphish also lacks the Whatsapp phishing simulations, considering Whatsapp as a very common attack vector in countries like Pakistan. The platform does not include features like AI-powered adaptive content, real-time risk-based learning, and offers no exclusively tailored support for diverse groups like students and general public users. [6]

- **TryHackMe:** TryHackMe is a renowned hands-on cybersecurity training platform that gamifies the learning process through virtual labs, interactive challenges, and competitions such as "King of the Hill." Users learn by solving hands-on hacking scenarios in isolated environments while proceeding through a structured learning path. Technical skill development is their primary focus for persons interested in ethical hacking and penetration testing. TryHackMe does not offer phishing campaign simulations, AI-driven risk analytics, multilingual accessibility, or administrative features for multi-level group monitoring and group onboarding. [7]

- **GoPhish:** GoPhish is an open-source simulation tool for phishing that is widely utilized for running email phishing campaigns. With GoPhish, administrators can create mail templates, landing pages, and manage user groups, hence offering basic campaign management and reporting features. Although it is an impressive tool for testing the anti-phishing strength in an organization, it does not have any such comprehensive training features like structured courses, quizzes or certifications, etc. Also, there is no in-built adaptive risk scoring, multilingual support, or LMS-style progress tracking.[8]

- **Guardey:** Guardey serves gamified cybersecurity awareness training designed for short engaging challenges mimicking the feel of apps like Duolingo. It incorporates narrative, leaderboards, and badges to keep things fun and entertaining for the users, while administrators find value in real-time reporting and analytics. In addition, phishing and spear-phishing simulations are provided to reinforce the awareness of users. However, it does not provide features like personalizing the training content based on individual user performance and providing training content more adaptive to Pakistani context. [9]

- **CybeReady:** CybeReady delivers context-aware personalized training via a smart adaptive learning engine, a product of machine learning with its data trained on an assortment of numerous cybersecurity threats for mastering through training. It delivers phishing and smishing simulations in multilingual format. The key features made available are security bites, training decks, and live phishing reporting with PhishCage. These features, however, place CybeReady mainly in the enterprise and employees settings, with no flexibility to serve educational institutions or general learners. It does not include features like voice phishing simulation, or advanced role-based group management. [10]

### Gaps in Existing Systems

- They are designed mainly for corporate enterprises and create barriers for students and non-affiliated users.
- These systems lack crucial support for regional languages like Urdu, and their training examples are based on foreign companies, rendering the content irrelevant for a Pakistani audience.
- These solutions mainly deal with email phishing, while completely ignoring other attack vectors like Whatsapp phishing which is a popular means for phishing attacks in many countries including Pakistan.
- The assessment methods generally are by way of simple multiple-choice or true-false questions that do not entirely probe the extent of a user's knowledge.
- Administrative features in existing platforms are not flexible and do not provide the granularity required for the hierarchical management of institutions.
- The existing systems lack the integration of AI for generating quizzes and phishing content.

## Features of Proposed System

### Platform Roles

- **System Administrator:** They have complete control over the entire system.
- **Client Administrator:** They manage one specific educational institution along with that institution's hierarchical user groups.
- **Affiliated User:** These are individual learners who are members of an institution administered by a client administrator (i.e. student).
- **Non-Affiliated User:** They are individual learners not part of any institution and can only access the information that is made accessible to the general public.

**Features**

**Authentication & Onboarding**

- Secure user authentication will be provided which includes support for social logins, and session management.
- **Role-Based Onboarding**
    - **System Administrator:** They are onboarded manually at first platform setup and can create institutional groups and assign Client Administrators.
    - **Client Administrator:** They are onboarded by a System Admin. They receive the rights to manage a specific institution including creating sub-groups within the institution and onboarding users.
    - **Affiliated User:** They are bulk onboarded. The Client Administrator uploads a user list (e.g. student emails), after which invitation emails are sent to users for easy signing up.
    - **Non-Affiliated User:** They are self-onboarded by registering themselves at the sign-up page.

**Learning & Training Features**

- **Role-Based Access & Content Management**
    - **System Administrator:** They can upload global courses and videos that are accessible from all user groups. Has full CRUD access on all content and learning paths.They can use AI to generate quizzes from uploaded text/PDFs (domain knowledge) or create them manually.
    - **Client Administrator:** They can upload and manage courses and videos strictly within their user groups and subgroups.
    - **Affiliated User:** They can view courses, videos, and quizzes assigned by their Client Administrator.
    - **Non-Affiliated User:** They have default access to general cybersecurity quizzes and related content.
- **Multilingual Monitored Courses**

    All roles interface the entire platform and content in English and Urdu, supported by Google Translate API. Courses include videos and quizzes, leading to the issuance of certificates.
- **Reporting**
    - **Affiliated & Non-Affiliated Users:** They can view their own quiz scores and progress report.
    - **System & Client Administrators:** They can generate full detailed

reports on user progress, quiz scores, and course completion depending on their domain (global vs. their group).

**Phishing Simulation Features**

- **Multi-Vector Campaign Management**
    - **System Administrator:** They can create and manage advanced phishing campaigns (email, Whatsapp-smishing, and browser-based voice vishing) against all users, including non-affiliated users. AI-generated customizable text for email & Whatsapp phishing and fake landing pages are used for those campaigns. It is possible to launch targeted campaigns according to user risk levels.
    - **Client Administrator:** They can create and run phishing campaigns only for their own user groups and sub-groups.
    - **Affiliated & Non-Affiliated User:** They receive a simulated phishing campaign as part of their training.
- **Real-Time Phishing Analysis**
    - **Email/Whatsapp Analysis:** Based on whether the user clicks on the simulated phishing link and interacts with it.
    - **Voice Vishing Analysis:** AI sentiment analysis is applied to the call transcript to verify if the user was scammed.
- **Reporting**
    - **System Administrators:** They can generate extensive reports for all phishing campaigns throughout the platform, including click-through-rate, and metrics on user engagement.
    - **Client Administrators:** They can generate reports about the phishing campaigns within their group.
    - **Affiliated & Non-Affiliated Users:** They can generate reports about their performance in the phishing campaigns.

**Unified Risk Analysis**

- The system will calculate a unified risk score for each user by combining their Phishing simulation performance (link clicks, failed phishing calls) with their LMS performance (course completion, quiz scores)
- **System & Client Administrators:** They are capable of monitoring aggregated risk levels, click rates, and user participation via comprehensive dashboards; they are able to view users sorted by risk level.
- **Affiliated & Non-Affiliated Users:** They would be allowed to see their own risk score and progress dashboards as well.
- This system automatically assigns remedial courses to users failing a phishing simulation (high risk), and once the course is completed, the risk score will be

reconsidered.

**Gamification Features**

- **Points & Leaderboards**
    - Points may be added or deducted based on performance in LMS and phishing attempts.
    - **System Administrator:** They can see a global leaderboard across all platforms and group-specific leaderboards.
    - **Client Administrator:** They can see the leaderboard of their own group.
    - **Affiliated User:** They can see the leaderboard of the group to which they belong.
    - **Non-Affiliated User:** They can see the global leaderboard.
- **Badges & Certificates:** They are awarded to users for course completion and good performance in simulations, which are displayed on user profile.

**Simulated Incident Reporting Features**

- **Affiliated & Non-Affiliated Users:** They can submit a report through a form with fields like category (Whatsapp/Gmail/Voice), sender, subject, and content etc.
- The system verifies the reports as Correct (True Positive) or Incorrect (False Positive) by matching it against the backend database of simulated threats. The system updates the risk Score based on accuracy and reporting delays (Time to Report).
- **System & Client Administrators:** This data is then represented on the system and client administrator's dashboards.

**Communication & Support**

- All users receive system notifications
- An AI chatbot provides 24/7 support.

## Project Overview/Goal

CyberShield Awareness Portal **aims** to create a singular system of multilingual cybersecurity training for educational institutions and the general public, who have been denied training owing to enterprise-driven solutions. The platform **solves** the problems with practical preparedness that is regionally relevant with respect to threats such as Whatsapp and voice phishing by integrating an LMS with AI-driven advanced simulation training in English and Urdu.Its **key benefits** include practical engaging training through gamification with realistic multi-vector phishing campaigns (email, Whatsapp, voice) and granular risk analytics for administration. The **final product** will be a web app built on a Next.js/Node.js stack and MongoDB that will incorporate AI APIs for content generation and require no special hardware for access via any browser.

## Alignment with UN Sustainable Development Goals (SDGs)

- **SDG 4: Quality Education:** It aims to follow it by providing accessible multilingual cybersecurity education for institutions and the public.
- **SDG 9: Industry, Innovation, and Infrastructure:** It aims to follow it by building a resilient digital infrastructure through increased awareness of cybersecurity.
- **SDG 10: Reduced Inequalities:** The platform aims to provide inclusive, regionally relevant training.

## Project Development Methodology / Artitecture

### Approach

We will follow an Agile Incremental Model:

- **Incremental Delivery:** Features will be divided into modules (Learning & Simulation) and built in iterations.
- **Continuous Feedback:** Regular advisor check-ins to refine requirements.
- **Layered Design:** Each module will have presentation (UI), application (logic), and data (DB + AI APIs) layers to ensure scalability.

### Modules & Objectives

### Module A: Learning Management & Training

### Objectives

- Provide role-based, multilingual learning paths (English and Urdu) for different user types (Affiliated and Non-Affiliated Users).
- Support secure authentication and onboarding, including social logins, bulk onboarding for affiliated users, and self-registration for non-affiliated users.
- Allow content upload and management by System and Client Admins, with both manual and AI-assisted quiz generation from text/PDFs.
- Deliver videos, courses, and quizzes to learners, issuing certificates upon completion.
- Enable reporting of progress and quiz scores for learners, with detailed reports available to administrators.
- Motivate learners through gamification with points, badges, certificates, and leaderboards visible at global, group, and user levels.

- Provide system notifications and chatbot support to improve the user experience.

**Key deliverables**

- **Authentication & Onboarding:** Secure login with social logins, bulk onboarding for students, and self-registration for non-affiliated users.
- **User Interfaces:** Role-specific pages for admins and learners, accessible across devices in English and Urdu.
- **Course & Quiz System:** Tools for admins to upload courses and for learners to take quizzes, with AI-generated or manually created quiz content.
- **Certificate Generation:** Automatic issuance of certificates for course completion.
- **Reporting Tools:** Learners can view their own reports; admins can generate detailed progress and completion reports.
- **Gamification Features:** Points, badges, and leaderboards tied to both training and phishing performance.
- **Communication & Support:** System-wide notifications and an AI-powered chatbot for 24/7 assistance.

**Module B: Phishing Simulation & Unified Risk Analysis**

**Objectives**

- Provide multi-vector phishing simulations (Email, Whatsapp, and Voice) managed by System Admins (globally) and Client Admins (within groups).
- Use AI-generated phishing content for email and Whatsapp templates.
- Track user behavior during simulations, including submissions, and responses to voice calls.
- Perform real-time phishing analysis, applying sentiment analysis to vishing transcripts.
- Calculate a unified risk score by combining LMS performance (courses/quizzes) with phishing results.
- Automatically assign remedial courses to high-risk users and update scores after completion.
- Generate reports and dashboards for administrators to monitor risk levels, campaign effectiveness, and user participation.
- Allow learners to submit incident reports (suspected phishing emails, Whatsapp) and validate them against the system's database.

**Key deliverables**

- **Campaign Management:** Tools for admins to design, launch, and schedule phishing campaigns by email, Whatsapp, or voice.
- **AI-Generated Content:** Realistic phishing templates and scripts automatically produced with AI.
- **Simulation Tracking:** Capture user interactions (submissions, call outcomes, transcripts).
- **Transcript & Sentiment Analysis:** Analyze vishing call transcripts to assess user responses.
- **Unified Risk Scoring:** Automated scoring that combines LMS and phishing results, updating user dashboards and triggering remedial courses.
- **Admin Dashboards & Reports:** Visual insights into risk levels, and engagement, with downloadable summaries.
- **User Risk Dashboards:** Learners can view their own risk score and progress reports.
- **Incident Reporting System:** Learners can report suspicious messages or emails, with system verification of true/false positives and score updates.

**Tools & Justification**

**Frontend (User Interfaces)**

- **Next.js / TypeScript**: Build responsive, role-based dashboards and training pages with fast rendering and mobile-friendly design.
- **Clerk**: Provide secure login and role-based access.

**Backend (APIs & Logic)**

- **Node.js (Express)**: Core APIs for courses, quizzes, campaigns, and risk scoring.
- **Google Translate API**: Provides multilingual support (English and Urdu) across the platform.

**Data Management**

- **MongoDB Atlas**: Stores user data, progress, transcripts, and risk scores with flexible schemas for supporting institutional hierarchies (system admin → client admin → groups → users).
- **Cloudinary**: Media storage for training videos, images, and certificates.

**AI & Telephony Services**

- **Gemini API**: Generates phishing email content and quizzes from PDFs/text.
- **ElevenLabs**: Simulate voice phishing calls (vishing), return transcripts, and enable sentiment analysis.

- **Twilio**: Whatsapp phishing simulations.
- **Dialogflow ES**: Provides 24/7 learner support through live chat and automated responses.

**Cloud Deployment**

- **Vercel**: Hosts frontend and backend on a serverless environment with automatic deployments.

**Team Operations & Collaboration Tools**

- **GitHub**: Source control, pull requests, and CI/CD pipelines.
- **Linear**: Sprint planning and task tracking.
- **Slack**: Team communication.
- **Google Drive**: Project documentation and notes.
- **Figma**: Wireframes and interface design for frontend.

**Note:** System Level Block Diagram is Present at the End of Document.

## Project Milestones and Deliverables

| **Name** | **Start** | **Finish** | **Duration** |
|---|---|---|---|
| **Cybersecurity Awareness Platform (Overall Project)** | 01/09/2025 | 11/05/2026 | 181 day |
| **Requirement Analysis** | 01/09/2025 | 22/09/2025 | 16 day |
| Requirement Research (study existing tools, needfinding target users, finalize feature list) | 01/09/2025 | 15/09/2025 | 11 day |
| SRS Document | 16/09/2025 | 23/09/2025 | 6 day |
| **Design** | 24/09/2025 | 22/10/2025 | 21 day |
| System Architecture Style & Block Diagram | 22/09/2025 | 29/09/2025 | 6 day |

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|---|---|---|---|
| Database Schema & API Design | 30/09/2025 | 14/10/2025 | 11 day |
| UI/UX Prototypes (wireframes, dashboard layouts) | 15/10/2025 | 22/10/2025 | 6 day |
| **Development** | 20/10/2025 | 20/03/2026 | 110 day |
| Database Development (MongoDB setup, schema) | 20/10/2025 | 10/11/2025 | 16 day |
| **Front-End Development** | 11/11/2025 | 09/01/2026 | 44 day |
| Learning Management (courses, quizzes, certificates) & Reporting frontend | 11/11/2025 | 10/12/2025 | 22 day |
| Dashboard & Gamification (leaderboards, points) & Reporting frontend | 11/12/2025 | 09/01/2026 | 22 day |
| **Back-End Development** | 09/01/2026 | 31/03/2026 | 58 day |
| Phishing Simulation Engine (email, Whatsapp messaging, voice) + Reporting backend | 09/01/2026 | 10/02/2026 | 23 day |
| AI Integration (Gemini content gen, ElevenLabs voice) +Reporting backend | 11/02/2026 | 02/03/2026 | 14 day |
| Risk Scoring & Analytics Engine & Reporting backend | 03/03/2026 | 23/03/2026 | 15 day |
| System Integration (frontend + backend + AI APIs) | 24/03/2026 | 02/04/2026 | 8 day |

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|---|---|---|---|
| **Testing & Deployment** | 01/04/2026 | 30/04/2026 | 22 day |
| Software Testing (unit, integration, user) | 01/04/2026 | 20/04/2026 | 14 day |
| Implementation & Deployment (Vercel, MongoDB Atlas) | 21/04/2026 | 30/04/2026 | 8 day |
| **Final Documentation & Defense** | 21/04/2026 | 11/05/2026 | 15 day |

**Work Division**

**Maheen Akhtar Khan**

- Backend development (Node.js APIs, risk scoring engine, AI integrations with Gemini & ElevenLabs, database design).
- System integration (frontend + backend + AI services).
- Testing (backend APIs, risk analysis, phishing simulations).
- Documentation (system architecture, backend design, deployment details).

**Aimen Munawar**

- Frontend development (Next.js dashboards, gamification, reporting UI).
- Backend contributions (phishing simulation engine and reporting).
- Testing (UI/UX, frontend functionality, user dashboards).
- Documentation (UI/UX design, frontend modules, test reports).

**Hadia Ali**

- Frontend development (LMS features: courses, quizzes, certificates UI).
- Backend contributions (campaign management, reporting, MongoDB integration).
- Testing (UI/UX, LMS workflows, certificates, onboarding flows).
- Documentation (UI/UX design,requirements analysis, feature list, project report assembly).

**Costing**

| Tool / Service | Usage in Project | Plan Used | Estimated Cost |
|---|---|---|---|

| | | | |
|---|---|---|---|
| **Next.js, TypeScript, Tailwind** | Frontend framework, styling, and UI components | Open-source, free | **0 PKR** |
| **Clerk** | Authentication with role-based access and social logins | Free plan (10k monthly active users) | **0 PKR** |
| **Node.js (Express)** | Backend APIs and logic | Open-source, free | **0 PKR** |
| **Google Translate API** | Multilingual course support (English ↔ Urdu) | Basic Tier (first 500,000 characters per month are free) | **0 PKR** |
| **MongoDB Atlas** | Database for users, progress, campaigns, transcripts storage | Free tier (512 MB storage, shared cluster) | **0 PKR** |
| **Cloudinary** | Media storage for videos, images, certificates | Free plan (25 monthly credits) | **0 PKR** |
| **Gemini API** | AI-generated quizzes, phishing content | Free quota (15 req/min, 1M tokens/month) | **0 PKR** |
| **ElevenLabs** | Voice phishing simulation | Free plan (on generic LLM) | **0 PKR** |
| **Twilio** | WhatsApp phishing simulations | Free (Meta's marketing template) | **0 PKR** |
| **Dialogflow ES** | Chatbot endpoint for 24/7 learner support | Trial Edition (Text queries, knowledge connectors free) | **0 PKR** |
| **Vercel** | Hosting for frontend + serverless backend APIs | Free plan (Hobby tier: 360 GB-hrs of provisioned memory) | **0 PKR** |
| **GitHub** | Source control, CI/CD pipelines | Free plan (Unlimited public/private repos) | **0 PKR** |
| **Linear** | Task and sprint tracking | Free plan (Unlimited members, 250 issues) | **0 PKR** |

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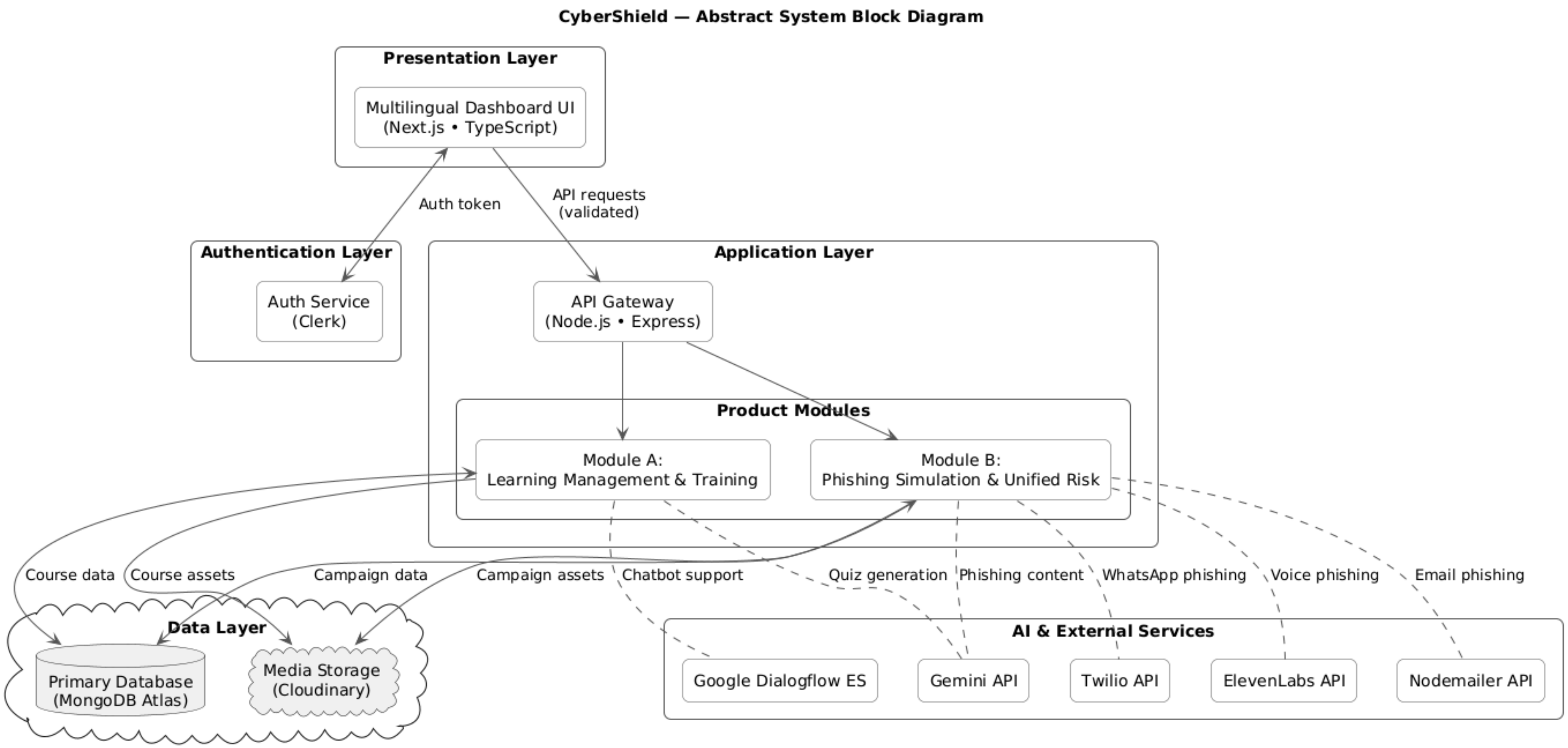
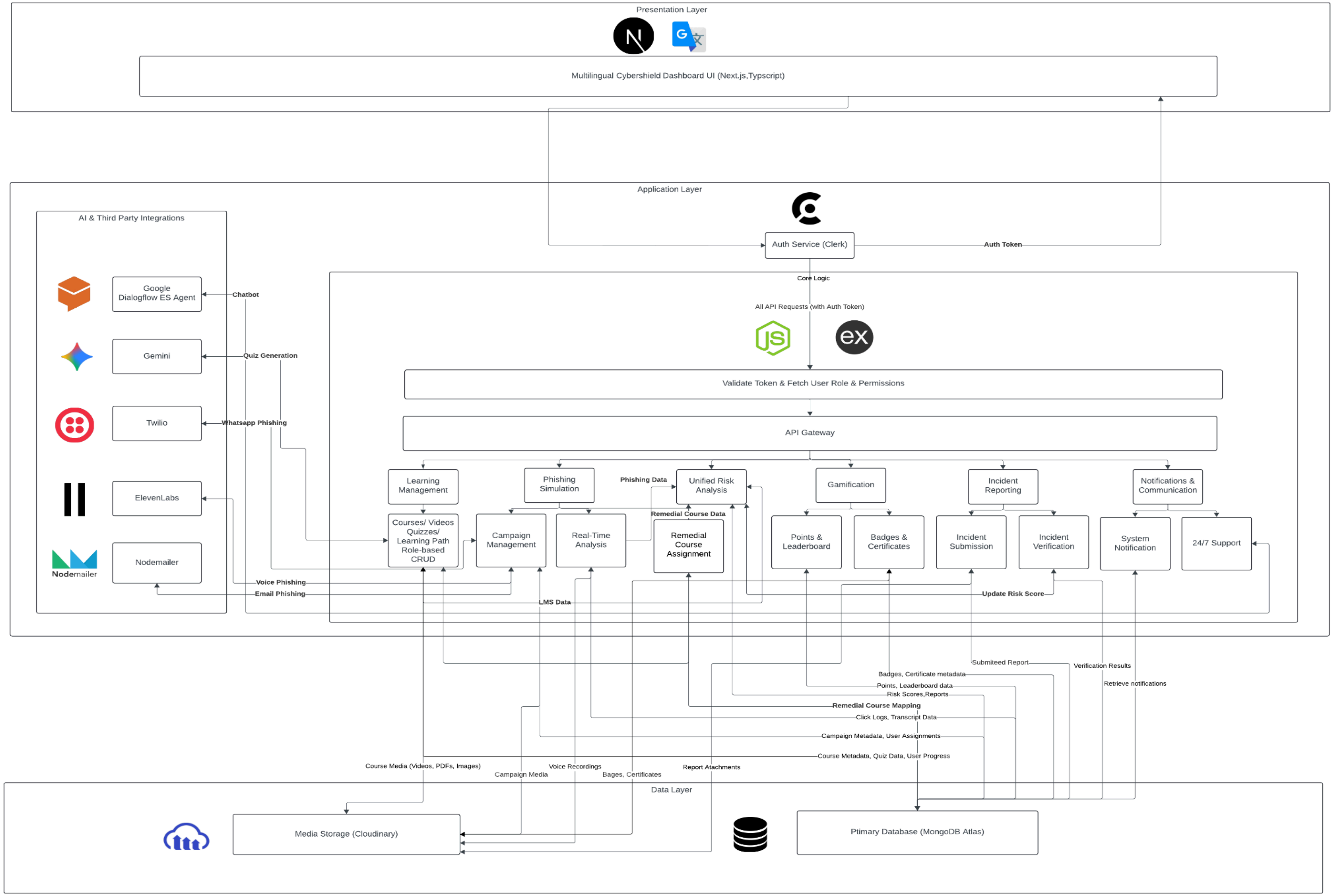
|---|---|---|---|
| **Slack** | Team communication | Free plan (message history up to 90 days) | **0 PKR** |
| **Google Drive** | Documentation and file storage | Free (15 GB) | **0 PKR** |
| **Figma** | Wireframes and UI design | Free plan (3 Figma files, collaborative editing) | **0 PKR** |

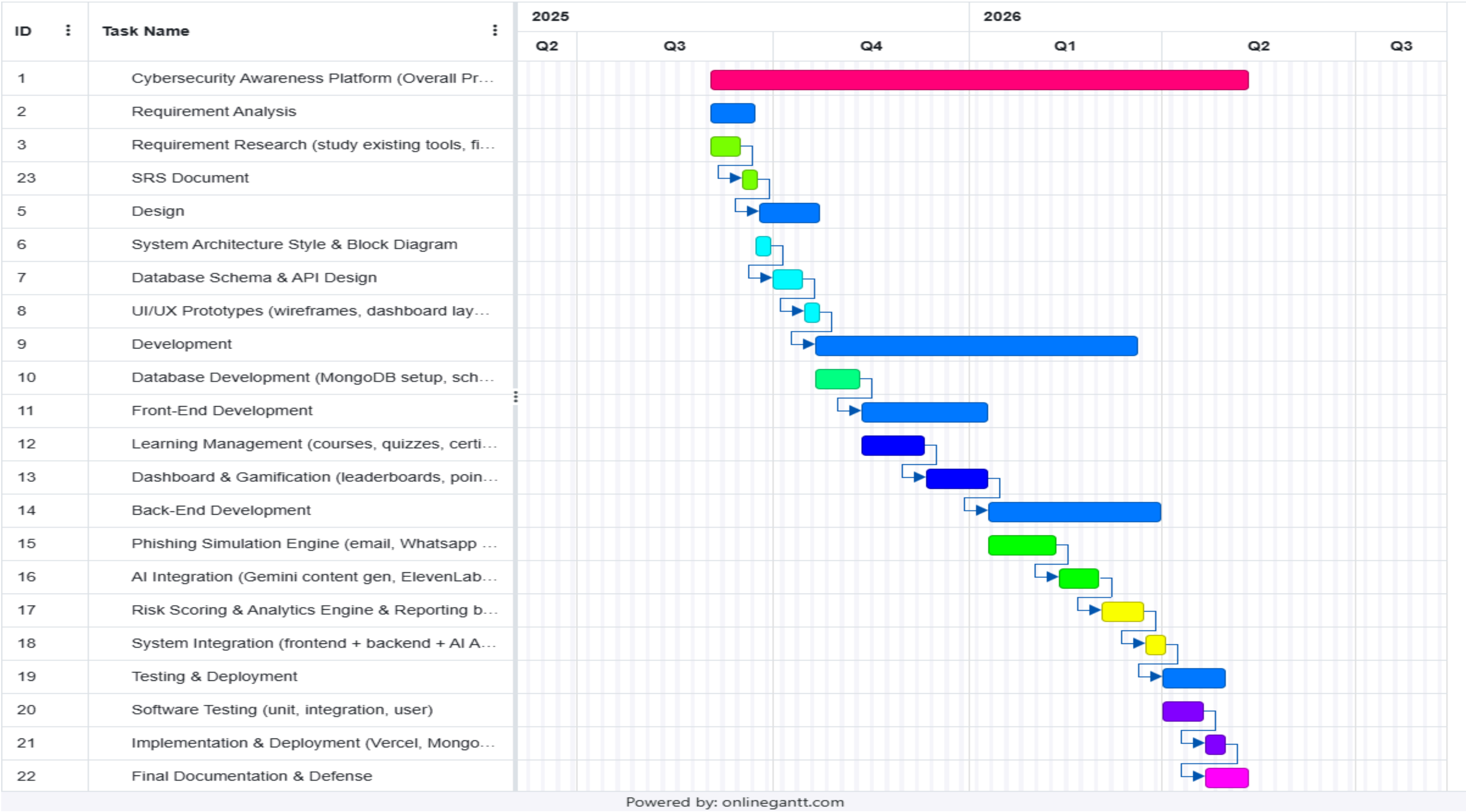
## Abstract System Level Block Diagram



## System Level Block Diagram



## Gantt chart


ID
Task Name
2025
2026
Q2
Q3
Q4
Q1
Q2
Q3
1 Cybersecurity Awareness Platform (Overall Pr…
2 Requirement Analysis
3 Requirement Research (study existing tools, fi…
23 SRS Document
5 Design
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7 Database Schema & API Design
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18 System Integration (frontend + backend + AI A…
19 Testing & Deployment
20 Software Testing (unit, integration, user)
21 Implementation & Deployment (Vercel, Mongo…
22 Final Documentation & Defense
Powered by: onlinegantt.com

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